Genevois C, Berthier P, Guidou V, Muller F, Thiebault B, Rogowski I. Effects of 6-Week Sling-Based Training of the External Rotator Muscles on the Shoulder Profile in High-School Elite Female Handball Players. J Sport Rehabil. 2014 Jul 23. [Epub ahead of print]

(Abstract)

CONTEXT: In female handball, the large numbers of throws and passes make the shoulder region vulnerable to overuse injuries. Repetitive throwing motions generate imbalance between shoulder internal and external rotator muscles. It has not yet been established whether sling-based training can improve shoulder external rotator muscle strength.

OBJECTIVE: This study investigated the effectiveness of a six-week strengthening program on improvement of shoulder functional profile in high school elite female handball players.

DESIGN: Crossover study.

SETTING: National elite handball training center.

PARTICIPANTS: Twenty-five high school elite female handball players.

INTERVENTIONS: The program, completed twice per week for six weeks, included sling-based strengthening exercises using a suspension trainer for external rotation with scapular retraction and scapular retraction.

MAIN OUTCOMES: Maximal shoulder external and internal rotation strength, shoulder external and internal rotation range of motion and maximal throwing velocity were assessed pre-intervention and post-intervention for dominant and non-dominant sides.

RESULTS: After sling training, external and internal rotation strengths increased significantly for both sides ($p \leq 0.001$, and $p=0.004$, respectively), with the result that there was no significant change in external and internal rotation strength ratios for either the dominant or the nondominant shoulder. No significant differences were observed for external rotation range of motion, while internal rotation range of motion decreased moderately, in particular in the dominant shoulder ($p=0.005$). Maximal throwing velocity remained constant for the dominant arm, whereas a significant increase was found for nondominant arm ($p=0.017$).

CONCLUSIONS: This 6-week strengthening program was effective in improving shoulder external rotator muscle strength, but resulted in a decrease in the range of motion in shoulder internal rotation, while throwing velocity remained stable. Adding a stretching program to this type of sling-based training program might help avoid potential detrimental effects on shoulder range of motion.